

B Suraj Patra

B.Tech CSE Student | Software Engineering & Full-Stack Development | Open-Source Contributor
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Summary

A Computer Science undergraduate with hands-on involvement in large-scale engineering projects and production-grade full-stack systems. Demonstrated ownership through open-source contributions and real-world deployments, with a strong foundation in data structures and algorithms.

Skills

Technical Skills

- Programming: C, Java, JavaScript, SQL
- CS Fundamentals: Data Structures, Algorithms, Object-Oriented Programming
- Full-Stack: MERN (MongoDB, Express.js, React, Node.js), REST APIs, React Native
- DevOps & Tools: Git, GitHub, GitHub Actions, Docker

Soft Skills

- Problem Solving, Critical Thinking, Leadership, Collaboration, Agile/Scrum

Projects

KL Eats

- Architected a campus food-ordering platform for **4.9K+ users**, achieving **99.9% uptime** and sustaining peak loads of **63+ RPS** (8650+ requests) validated via Locust load testing.
- Optimized frontend performance, increasing Lighthouse scores from **59 to 86** and reducing First Contentful Paint from **1.7s to 0.6s** through Redis-backed session management and UI restructuring.
- Implemented automated finance reconciliation and Telegram-based order orchestration, processing **Rs.1.5L+ monthly transactions**; operating under Equitech Lab Pvt. Ltd.
- **Tech Stack:** Next.js 15, React 18, TypeScript, Node.js, Express.js, MySQL, Redis, Tailwind, shadcn/ui

FaceAttend

- Architected a privacy-first facial recognition system with a distributed architecture (Mobile Kiosk, ERP Portal, FaceNet microservice), achieving **95% accuracy** and **1-2s latency**.
- Optimized performance by reducing inference calls by **60%** through on-device face pre-filtering and multi-embedding rotation (5 per user) using cosine similarity matching.
- Implemented secure device trust, audit logging, and **Android Lock Task Mode** to ensure system reliability and tamper-proof authentication; developed with Creovsys Solutions India Pvt. Ltd.
- **Tech Stack:** React Native (Expo), React (Vite), TypeScript, Node.js, Express.js, MongoDB, Python, Flask, FaceNet, Socket.IO, JWT

Education

K L E F Deemed To Be University – B.Tech CSE (2023–2027) — GPA: 9.40/10 — Vijayawada, India
Sunabeda Public School – Class XII (ISC, 2021–2023) — GPA: 8.50/10 — Sunabeda, Odisha, India
Sunabeda Public School – Class X (ICSE, 2020–2021) — GPA: 9.35/10 — Sunabeda, Odisha, India

Certifications

MongoDB Associate Developer | AWS Cloud Practitioner | GitHub Foundations

Patents & Publications

- **Biometric-based Voting System with Multi-layer Authentication** — IN202541073801 (Aug 2025). Invented a multi layer biometric voter authentication system to prevent impersonation and improve election transparency.
- **DigiChit : Automated Chit Fund Processing System** — IN202541098116 (Nov 2025). Developed a chit fund processing system integrating rule-based mandate verification, fraud-detection logic, and compliance automation.

Achievements

- Competitive Programming: LeetCode (Rating: 1600+) | CodeChef (Rating: 1500+)
- **Hackathons & Competitions:** Finalist, Build for Bharat Hackathon; Semi-finalist, Code for Bharat Season 2; Top 15, CodeRush Competitive Programming Challenge (IIITM Gwalior) — Team Code4Cause.
- Contributed to 9+ open-source repositories and supported open-data initiatives across Wikipedia, OpenStreetMap, and Weblate for regional-language accessibility.

Leadership

- **Head, Wikipedia Club** — **GNU/Linux Users Group (GLUG)**: Leading open-data and Wikipedia contribution initiatives, mentoring students, and promoting open knowledge through structured community participation.